

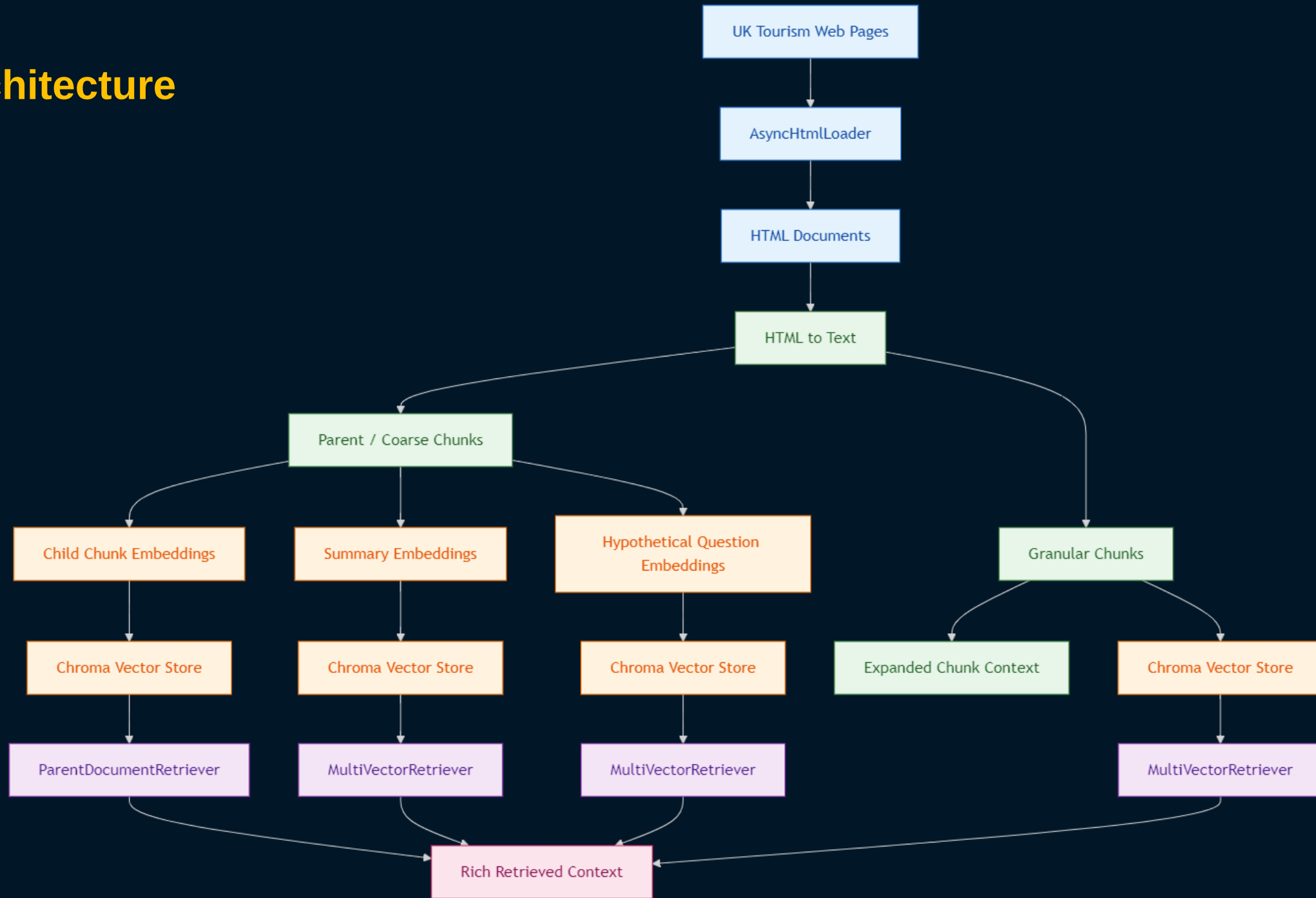
Agentic AI Engineering

Advanced Multi-Vector RAG Embedding Strategies with ChromaDB

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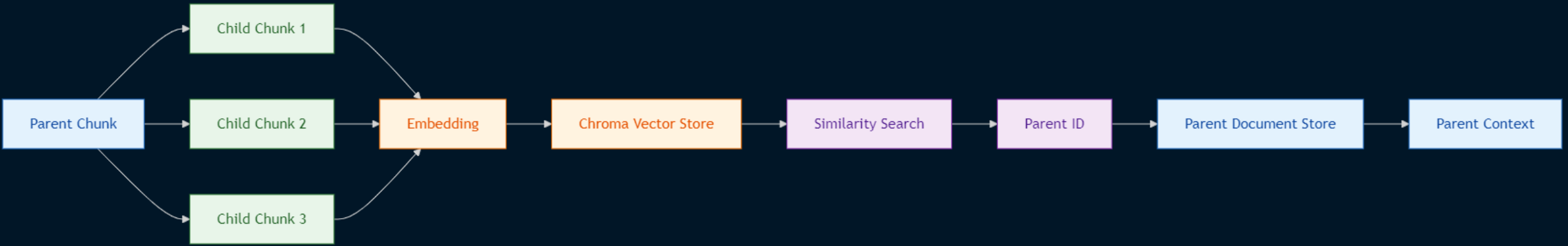
- Designed an **advanced multi-vector RAG architecture** separating retrieval representations from final answer context using LangChain ParentDocumentRetriever and MultiVectorRetriever.
- Implemented child-to-parent retrieval using granular 500-character chunks for semantic matching while returning larger 3,000-character parent contexts for downstream synthesis.
- Developed a **summary-embedding strategy** using LLM-generated summaries as retrieval vectors while preserving original coarse chunks as context.
- Implemented structured-output LLM workflows that generate exactly four hypothetical questions per document chunk to improve query-to-document semantic alignment.
- Built a **granular chunk-expansion strategy** that retrieves focused 500-character chunks while returning neighboring previous/current/next content for richer context.
- Integrated ChromaDB with OpenAI embeddings and in-memory document stores to support multiple vector-representation experiments across a multi-destination UK tourism corpus.
- Designed reusable indexing and retrieval components supporting comparative evaluation of child embeddings, summaries, hypothetical questions, parent documents, and expanded contexts.
- Established an extensible foundation for production RAG capabilities including persistent parent stores, hybrid retrieval, reranking, metadata filtering, retrieval evaluation, citation-aware generation, and observability

Architecture



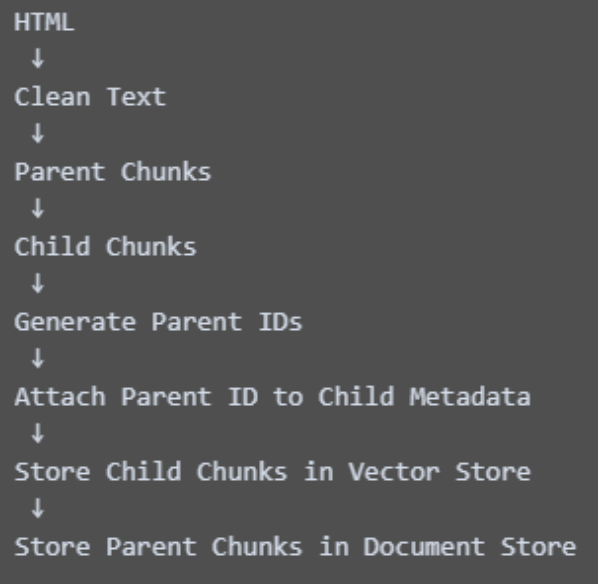
ParentDocumentRetriever Strategy

Data architecture

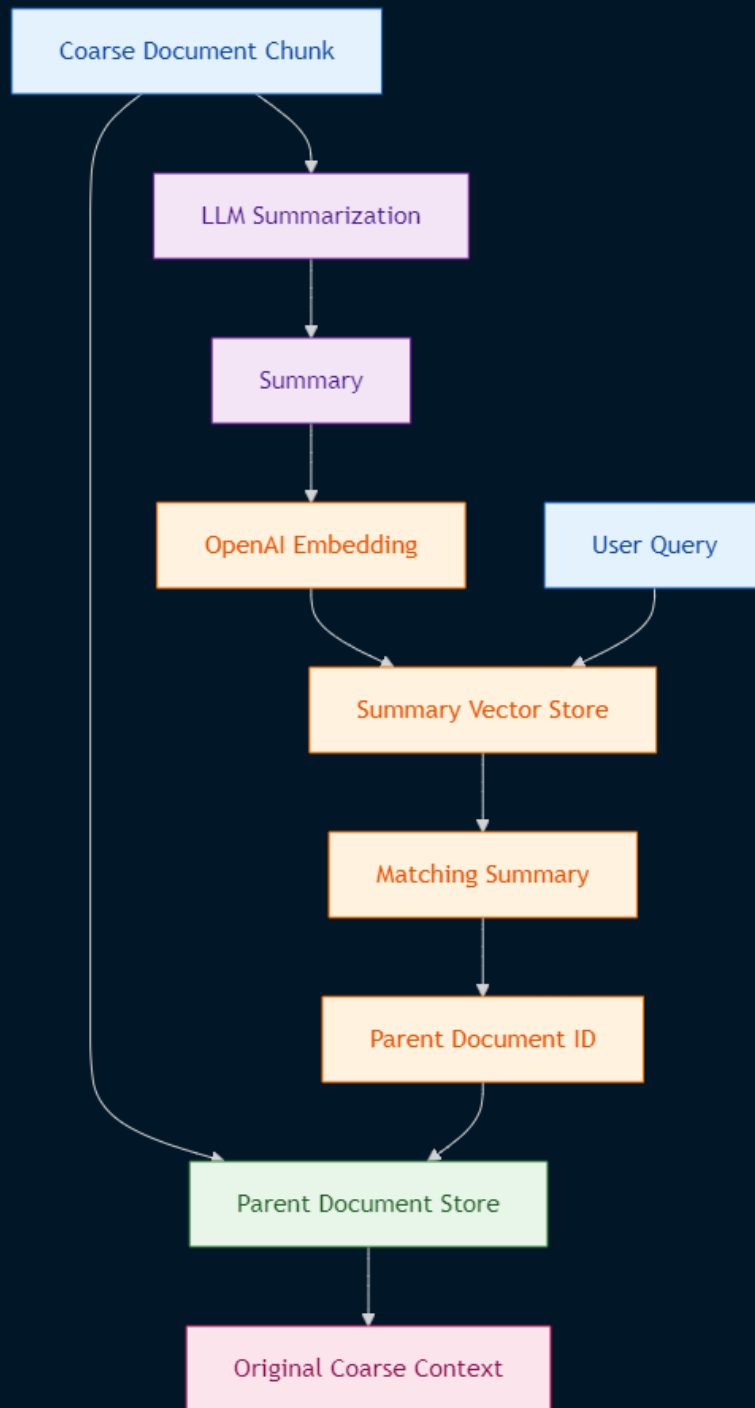


MultiVectorRetriever with Child Chunks Strategy

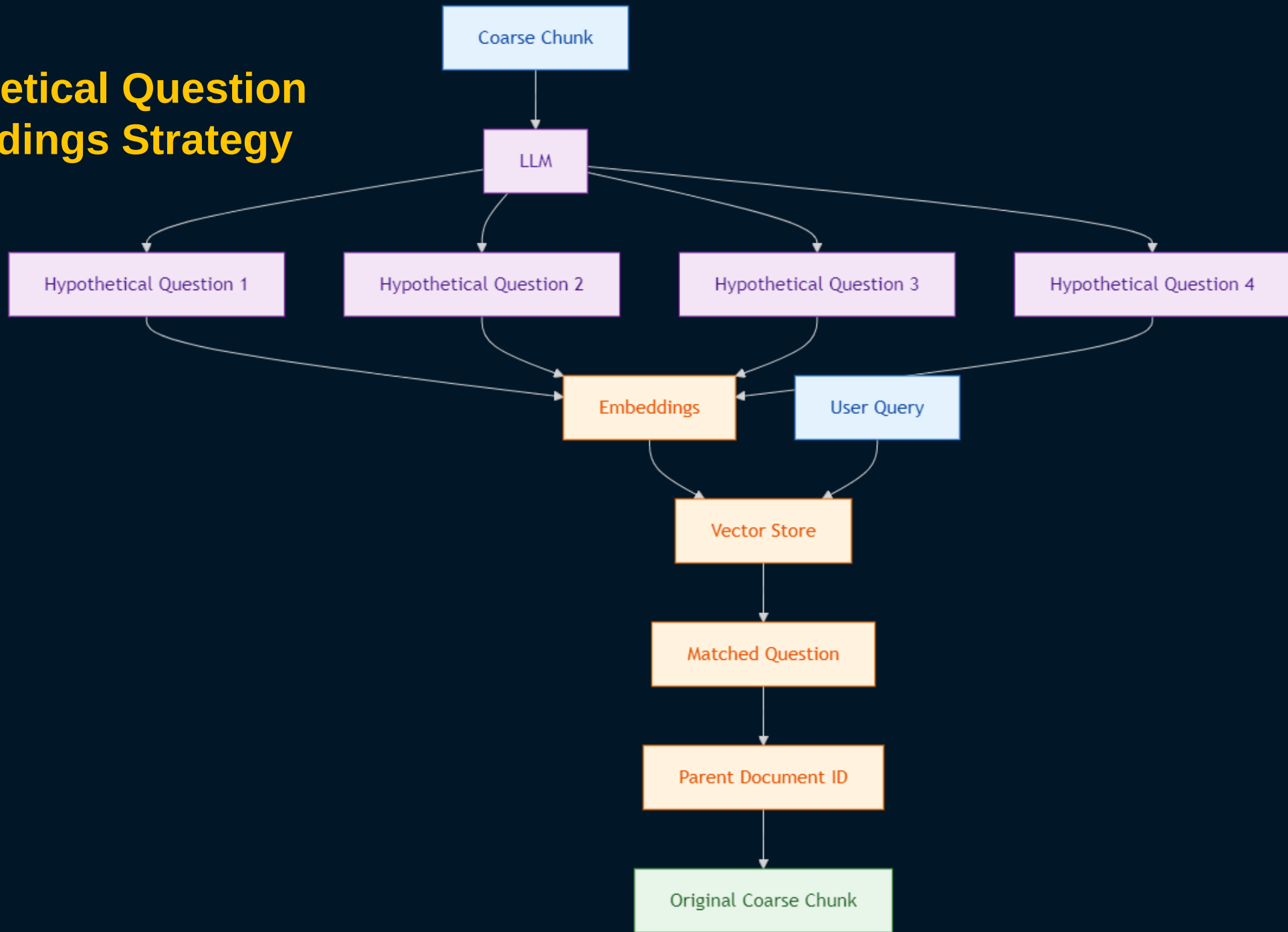
Ingestion flow



Summary Embeddings Strategy



Hypothetical Question Embeddings Strategy



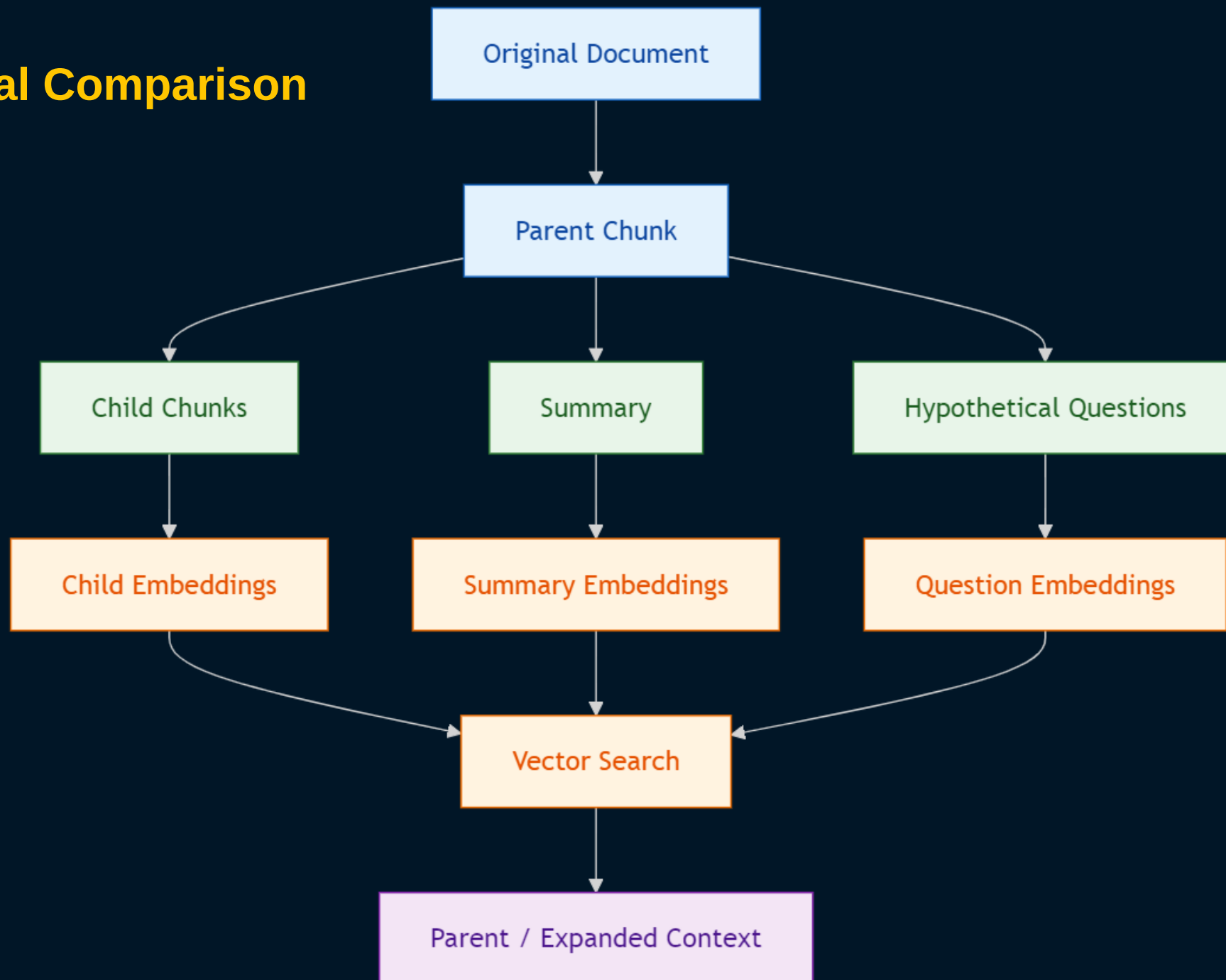
Granular Chunk Expansion Strategy



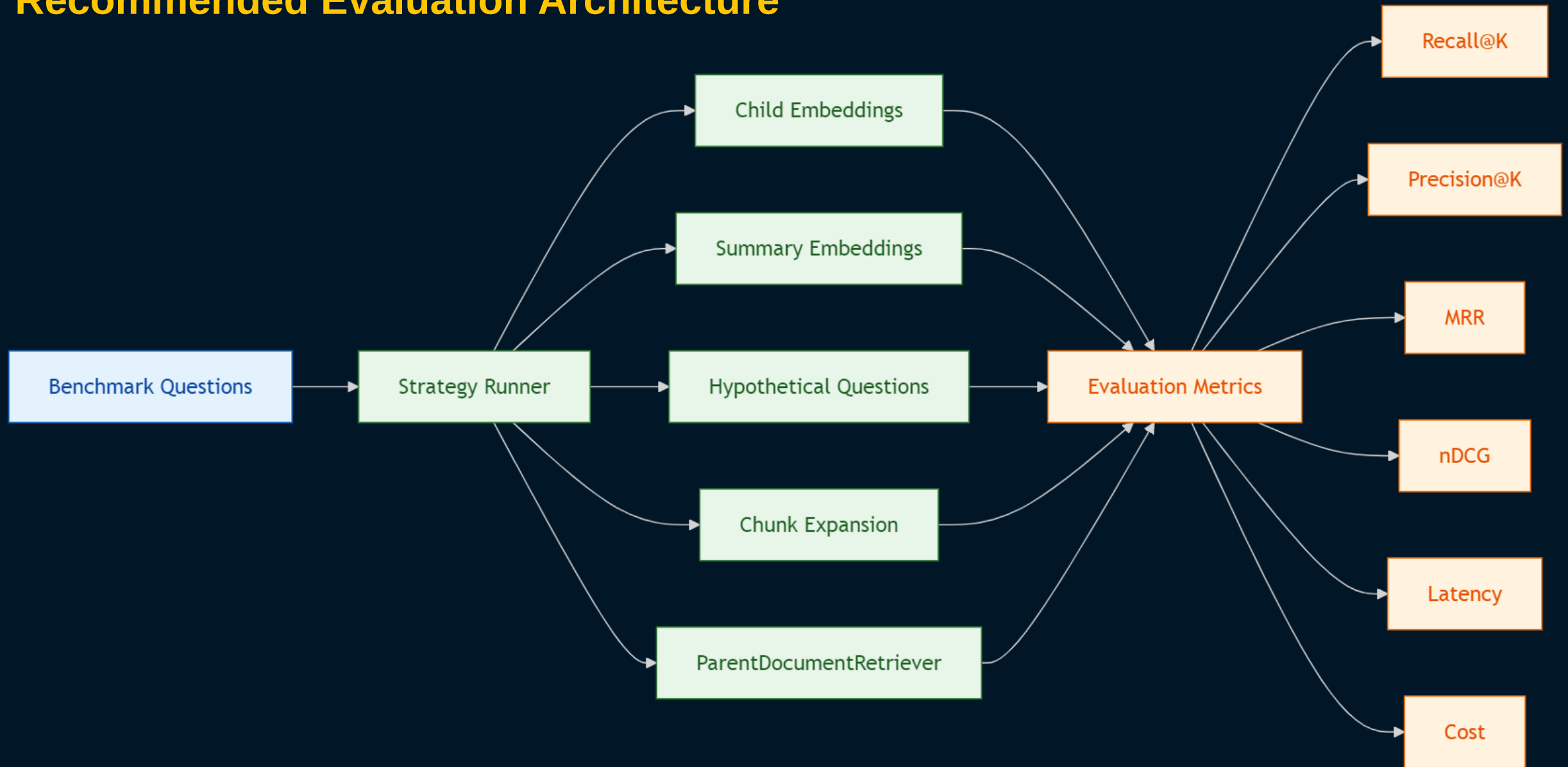
Strategy Comparison

Strategy	Retrieval Representation	Returned Context	Main Strength
ParentDocumentRetriever	Child chunks	Parent chunk	Precise retrieval + broader context
MultiVector + child	Child chunks	Parent chunk	Flexible multi-vector architecture
Summary embeddings	LLM summaries	Parent chunk	Focused semantic representation
Hypothetical questions	Generated questions	Parent chunk	Better query-to-document alignment
Chunk expansion	Current granular chunk	Previous + current + next	Local contextual enrichment

Conceptual Comparison



Recommended Evaluation Architecture



Enterprise RAG Architecture

